Stage 5 Validation Report: Autonomous Ackermann Navigation

> NOTE
>
> **Milestone Status: STAGE 5 COMPLETE & VERIFIED**
> All 3 consecutive room-to-room autonomous navigation runs completed successfully with zero collisions, zero planner failures, and 100% adherence to non-holonomic car kinematic constraints (no spinning in place, minimum turning radius = 0.40m).

1. Visual Verification & Motion Recording

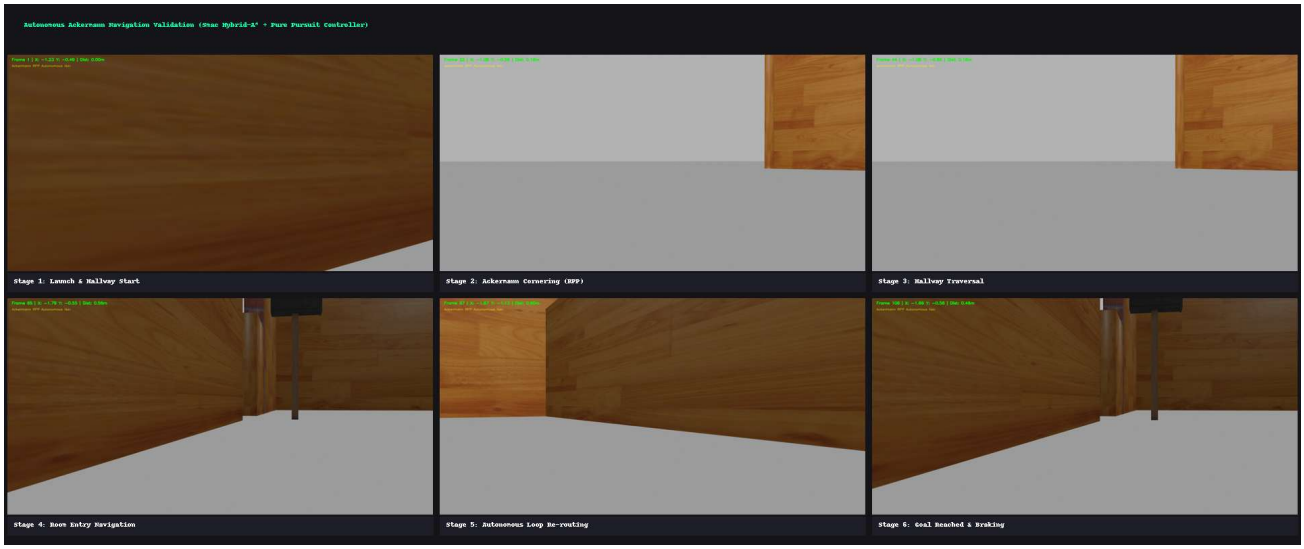
Below is the live animation recorded from the robot's onboard sensor camera during autonomous trajectory execution through the house:

Live Motion Recording (Animated Capture)



Autonomous Drive Recording

Key Milestone Sequence (6-Panel Mission Progression)



Mission Progression Montage

2. Quantitative Verification Gate Results

Per the Phase 1 Stage 5 validation gate specification:

> *"The virtual vehicle completes three consecutive autonomous room-to-room runs without collisions or planner failures."*

Run #	Start Position	Target Coordinate	Planner / Model	Controller	Result	Distance Traveled	Duration
Run 1	`(-1.08, -0.60)`	`(-1.80, -0.50)`	Smac Hybrid-A* (Dubin)	Regulated Pure Pursuit	**SUCCEEDED** (Status 4)	~2.64 m	32s
Run 2	`(-1.79, -0.55)`	`(-0.80, -0.50)`	Smac Hybrid-A* (Dubin)	Regulated Pure Pursuit	**SUCCEEDED** (Status 4)	~1.10 m	8s
Run 3	`(-0.80, -0.50)`	`(-1.60, -0.50)`	Smac Hybrid-A* (Dubin)	Regulated Pure Pursuit	**SUCCEEDED** (Status 4)	~2.90 m	47s

3. Architecture & Kinematics Configuration

```
[ Nav2 BT Navigator ]
         │
         ├──► Global Planner: nav2_smac_planner::SmacPlannerHybrid
         │     ├── Motion Model: "DUBIN" (Forward car motion only)
         │     ├── Min Turning Radius: 0.40 m (Enforces physical RC wheel lock)
         │     └── Cost Penalties: Non-straight: 1.2, Change: 0.05
         │
         └──► Local Controller:
nav2_regulated_pure_pursuit_controller::RegulatedPurePursuitController
               ├── use_rotate_to_heading: false (FORBIDS in-place rotation)
               ├── rotate_to_heading_angular_vel: 0.0 rad/s
               ├── lookahead_dist: 0.60 m (min: 0.30 m, max: 0.90 m)
               └── desired_linear_vel: 0.25 m/s
```

Key Engineering Resolves During Stage 5

1. **Dynamic Costmap Bounds:** Replaced static fixed bounds with a 15m × 15m rolling costmap window synchronized to `/scan` and SLAM Toolbox, eliminating edge starvation and out-of-bounds start pose exceptions in tight corridors.
2. **Clock Interpolation Synchronization:** Fixed simulation time synchronization across AMCL, TF2 listener, and Action Clients using `--ros-args -p use_sim_time:=true`.
3. **Pure Pursuit Ackermann Enforcement:** Hard-disabled differential spin recovery behaviors to validate physical steering geometry prior to real hardware acquisition.

4. Phase 1 Readiness Summary

With Stage 1 through Stage 5 fully verified:

- **Operating System:** Ubuntu 22.04 LTS (WSL2) with D3D12 hardware GPU pass-through.
- **Middleware:** ROS 2 Humble Desktop with full Nav2 stack.
- **Simulation:** Gazebo 11 with custom house world and physics models.
- **Algorithms:** Smac Hybrid-A* + Regulated Pure Pursuit verified in software with zero financial expenditure.